

The Coordinate Detector in GEp-V

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SBS Collaboration

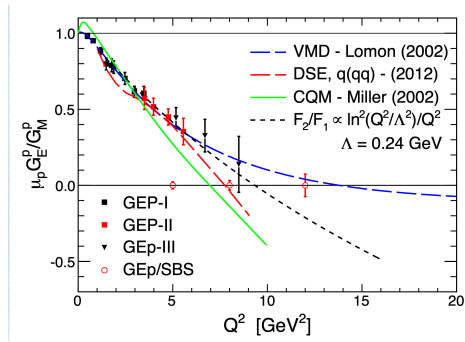
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GEP-V: the proton form-factor ratio at high Q^2

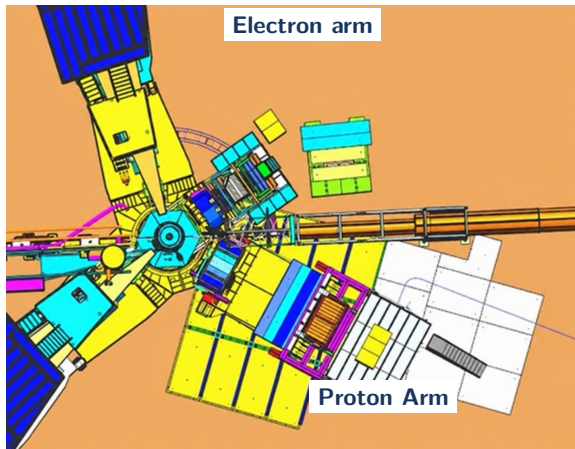
- ▶ GEp-V measures the proton elastic form-factor ratio G_E^p/G_M^p through recoil polarization.
- ▶ The high- Q^2 regime provides a sensitive test of QCD-inspired models of proton structure.
- ▶ The useful elastic signal is rare, and must be distinguished from both inelastic and electromagnetic background seen in the open-geometry apparatus.

Experimental requirement

Identify an electron and proton that satisfy the two-arm elastic kinematics, then measure the proton polarization.



GEp-V experimental layout



CDet links electron identification to the proton-arm tracking problem.

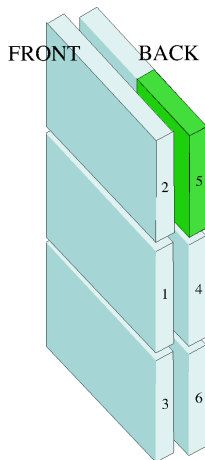
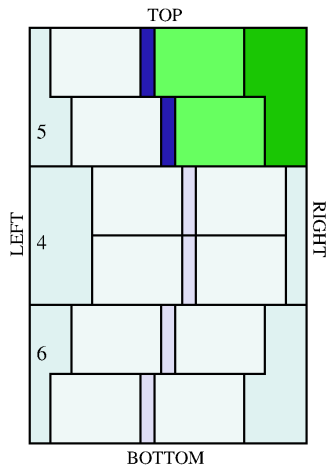
Elastic-event selection

- ▶ ECal measures the scattered-electron energy and position.
- ▶ Two CDet layers add a short tracking lever arm.
- ▶ The combined trajectory constrains the electron out-of-plane angle and the target association.

Front-tracker region of interest

- ▶ Electron-arm information predicts where the recoil proton should appear.
- ▶ A smaller GEM search region can reject unrelated hits.
- ▶ Reducing the combinatorics may also shorten reconstruction in the high-occupancy, open geometry.

CDet: two layers of segmented scintillator



- ▶ 168 scintillator bars in two layers
- ▶ 84 half-bars per layer
- ▶ Wavelength-shifting fibers carry light to multi-anode PMTs
- ▶ 16 pixels per half-bar give 2,688 logical channels

Measured quantities

Leading-edge time and trailing-edge time for each discriminator pulse.

The central challenge: hundreds of background hits

- ▶ Lower-energy electrons and photons reach CDet despite the aluminum shielding in front of the detector.
- ▶ At $I = 20 \mu\text{A}$ on a 15 cm liquid-hydrogen target, simulations and initial analysis indicate approximately
 - **500 hits/event** in Layer 1,
 - **250 hits/event** in Layer 2,within a 60 ns readout window around the coincidence-trigger timing signal.
- ▶ These occupancies remain after the hardware thresholds are applied.

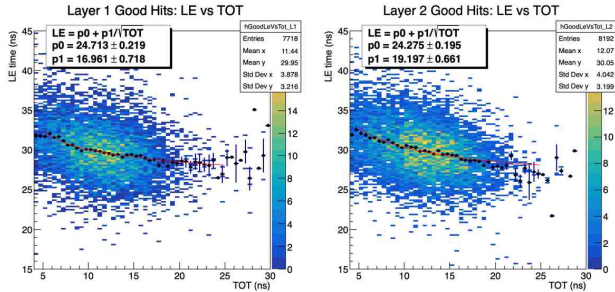
NINO threshold

The threshold corresponds to roughly 50% of the energy deposited by a minimum-ionizing, high-energy electron.

Analysis consequence

Timing alone cannot distinguish every signal hit from the large accidental population. Geometry and cross-detector consistency are essential.

TDC-only readout: time over threshold carries the charge information



Cross-target Run 3575: after channel-offset and ECal-time corrections, before time-walk correction

- ▶ Budget constraints precluded ADC instrumentation.
- ▶ The NINO pulse width provides time over threshold, a proxy for pulse amplitude.
- ▶ Larger pulses (larger ToT) cross threshold earlier. Here, earlier arrival means a smaller LE time, hence the downward slope.
- ▶ ToT supports a useful time-walk correction, but it does not replace a calibrated charge measurement.

The calibration must recover stable timing from leading edge and ToT alone.

Calibration strategy

Align the detector

- ▶ Extract channel-level timing offsets with half-bar information as a robust fallback.
- ▶ Correct common ECal-time dependence with a fixed-effects treatment across half-bars.
- ▶ Apply a layer-dependent ToT correction.

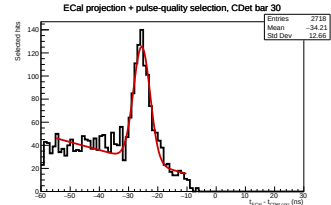
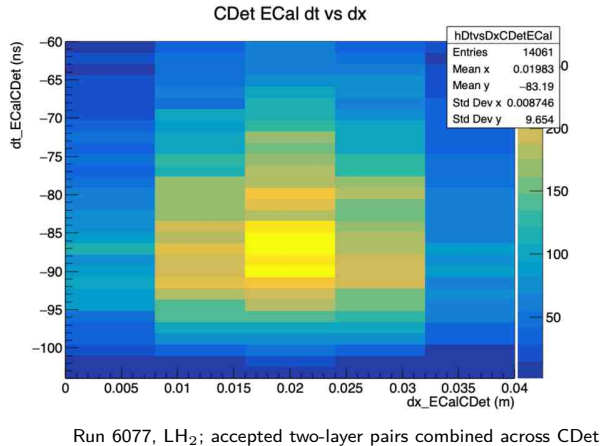
Build track candidates

- ▶ Form pulses from matched leading and trailing edges.
- ▶ Pair hits across the two CDet layers using timing and an ECal-projected trajectory.
- ▶ Retain exclusive single-layer candidates when one layer has no compatible hit.

Corrected timing observable

$$t_{\text{corr}} = t_{\text{LE}} + \Delta t_{\text{channel}} + \Delta t_{\text{ECal}} + \Delta t_{\text{ToT}} + \Delta t_{\text{run}}$$

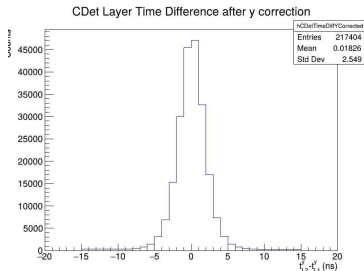
Corrected timing and trajectory identify high-energy electrons



Bar 30: ECal projection and pulse-quality selection

- ▶ High-energy electron candidates use ECal/HCal coincidence timing; the ECal position predicts the trajectory through CDet.
- ▶ Good CDet pairs form a compact locus in corrected time and Δx , rejecting the broad accidental background.

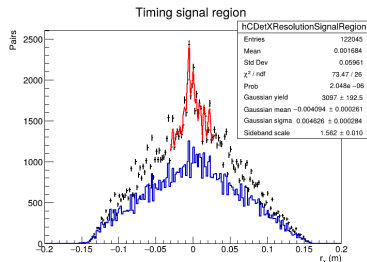
Current CDet timing and transverse-position resolution



$$\sigma(t_{L2} - t_{L1}) = 2.55 \text{ ns RMS}$$

Run 5710 cross-target; corrected accepted pairs

Equal, independent layers would give
 $\sigma_t \approx 2.55/\sqrt{2} = 1.80 \text{ ns per layer.}$

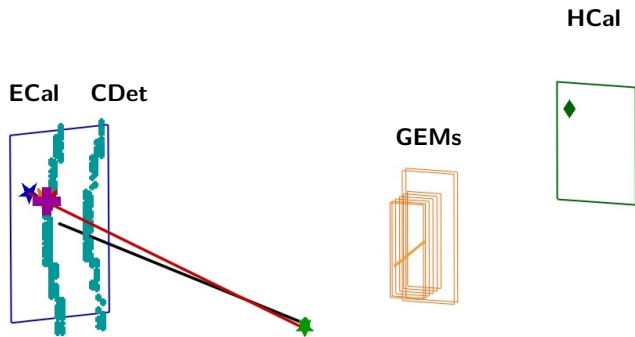


$$\sigma(r_x) = 4.63 \pm 0.28 \text{ mm}$$

Run 6077 LH₂; timing-sideband constrained fit

$r_x = (x_{L2} - x_{L1}) - (x_{\text{ECal}}/z_{\text{ECal}})\Delta z$. Equal,
 independent layers imply a preliminary $\sigma_x \approx 3.3$
 mm scale per layer.

CDet information now reaches the front-tracker ROI



- ▶ ECal and CDet define electron-ray hypotheses across the target.
- ▶ The replay exports ordinary pairs, seam pairs, and exclusive single-layer hypotheses.
- ▶ In a 261,010-event Run 5711 sample, 117 events contain both a front-GEM track and a CDet hypothesis.

Current status

The hypotheses support validation and diagnostics. They do not yet constrain production GEM pattern recognition.

Run 5711 event 38340, one of the 117 CDet–GEM overlap events

Status and next steps

Completed

- ▶ Timing calibration from TDC leading edge and ToT
- ▶ Detector-local pair and single-layer reconstruction
- ▶ ECal-projected trajectory matching
- ▶ CDet hypothesis export to FTROI

In progress

- ▶ Complete the target-coordinate transformation
- ▶ Validate elastic closure against reconstructed front-GEM tracks
- ▶ Quantify background rejection and GEM ROI reduction
- ▶ Measure any reconstruction-time improvement

Takeaway

CDet turns a very high-occupancy timing detector into usable electron-track constraints for GEp-V elastic-event reconstruction.